

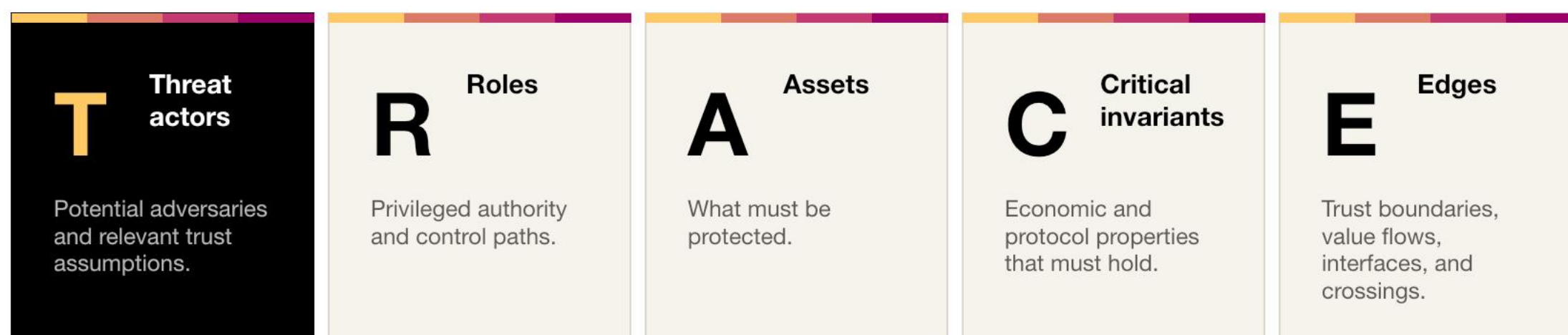
Oak TRACE Threat Modeling Framework

A Web3 design-risk method that turns heterogeneous protocol material into a structured model, then uses STRIDE, attack trees, and optional collusion analysis to produce decision-ready security work.

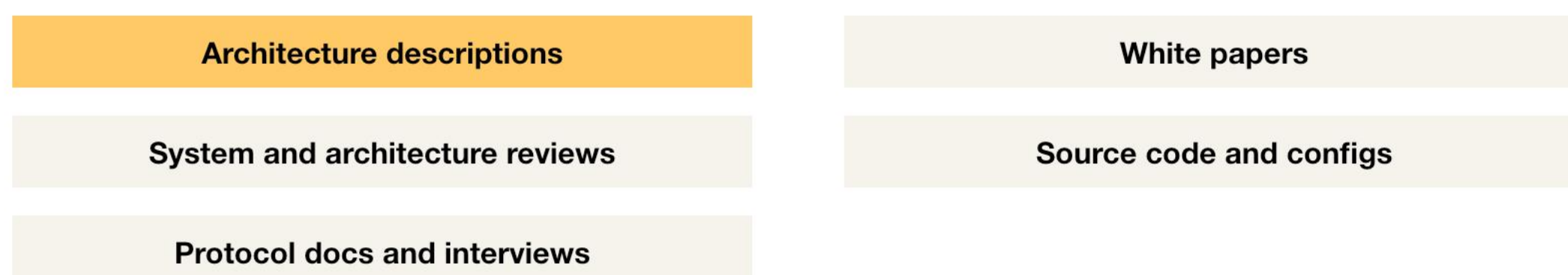
Why Web3 needs TRACE

Web3 projects combine high-value assets, economic invariants, off-chain systems, and human control paths. Oak TRACE keeps these in one model: STRIDE for conventional computing risk, plus attack trees and collusion/consensus analysis for protocol risk.

TRACE model primitives



Inputs



Output

A traceable model that connects each material threat to an asset, role, invariant, trust boundary, flow, attack tree, and mitigation decision.

From source material to attack paths.

AI accelerates decomposition and coverage. Oak senior security experts own the assumptions, severity, plausibility, and recommendations.

01 Ingest sources

Docs, reviews, architecture, white papers, code, configs, and protocol context.

02 Construct model

Assets, actors, invariants, boundaries, value flows, roles, components.

03 STRIDE analysis

Structured threat pass across components, flows, and privilege boundaries.

04 Attack trees

Top threats decomposed into plausible exploit paths and enabling conditions.

05 Collusion inspection

Optional pass for DAOs, validators, governance, committees, and multisigs.

06 Roadmap outputs

Risk ranking, mitigations, open questions, and audit or launch guidance.

Designed with human-AI co-working in mind

A deliberate sequential workflow uses AI for decomposition, coverage, and draft structure while Oak senior security experts own the risk decisions.

- AI proposes source inventories, model candidates, and coverage gaps.
- Oak experts approve and complement assumptions, scope, and relevance.
- Experts set severity, plausibility, and remediation guidance.

Human Decision Points

Source inventory, TRACE model, STRIDE Threat identification and ranking, attack trees, roadmap.